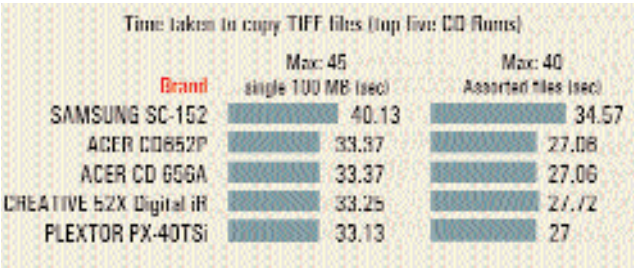


es from CDs), then the **ASUS CD-S520** is the best choice. For random data read applications, the **Apex CHI-56** gets top honours, but if features such as a remote that allows you to control drive operations



is more important to you, then the **Creative Digital iR** would be a good choice, provided you're willing to sacrifice on performance.

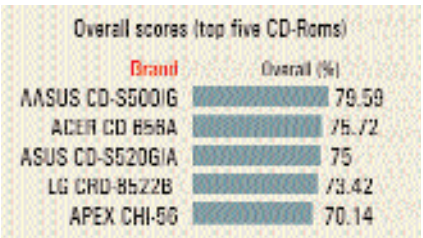
Audio ripping

This helps evaluate the ability of the drive to perform error corrections with the help of better cache synching on the fly, which results in WAV files devoid of any 'clicks' and jitters. This will result in better quality as well as faster conversion of CDA to WAV format.

All you MP3 buffs out there would need a drive that works well with applications that need audio tracks to be ripped from audio CDs so that they can be worked on and edited or converted to another format such as MP3. This process is called Digital Audio Extraction (DAE). This is a feature that older drives used to be incapable of supporting but is incorporated into all CD-ROM drives available today. Interestingly, this is also a feature that proves to be very well implemented in some drives and poorly implemented in others; although the data transfer might be fast in one, it could have very poor DAE speeds.

The drive that gave good results in this test was undoubtedly the **52x Acer CD652P**, which logged a blazing 16.94 seconds to rip the middle track—compared to the 43 seconds taken by the **ASUS S520** drive.

Another drive that also performed



very well in this test was the **Plextor PX-40Tsi**, taking just 27.03 seconds to rip the innermost track (72.84 MB in size) of the audio CD, making it almost twice as fast as the slowest performing drive in this test. Surprisingly, we found that the drive which excelled in data transfer—the **ASUS S520G 52x**—performed slowest in this test, taking a leisurely 73.68 seconds to rip the innermost track. This was almost three times more than the

fastest drive.

The **Samsung SC-152** also performed very well in this test, with very good results in spite of not having performed too well in the earlier data transfer tests.

So, if you are into converting music on CDs into MP3s or extracting from audio CDs, the best drive for you is the **Acer CD652P 52x**, which logged exemplary results in this test. Do keep in mind, however, that good performance in DAE and high data transfer performance cannot be directly co-related; there are drives that deliver very good data transfer performance, but lag in DAE performance.

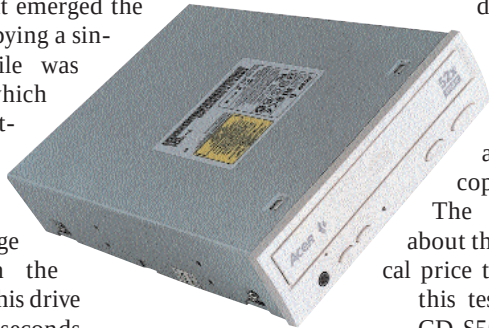
Working with files

Simply put, this is a set of tests that involves copying different file types from the CD-ROM to the hard disk. This is a very practical indication of the speed of the CD-ROM drive.

The drive that emerged the winner while copying a single 100 MB file was

Krypton 52x, which surprisingly outperformed all other drives in spite of delivering very average performance in the synthetic tests. This drive took only 26.19 seconds to copy the 100 MB test file to the hard drive. The scores in the synthetic tests for this drive were not among the highest, but as seen, it did very well in the real-world test. The runner-up in this test was the **52x LG CED 8120B**, which took 27.58 seconds to copy the file to the hard drive.

In the assorted file copy test, where an



Acer CD 652P: this drive gave above average results in most tests

CLV vs CAV

CLV (Constant Linear Velocity) is the method used to access data in almost all CD-ROM drives available in the market today. In this method the variable speed motor is used to spin the disc. The reason for this is that while reading the data available on the innermost track, whose circumference is smaller, the speed at which the disk spins has to be higher. On the other hand, when the head is reading the data from the outermost track, the rotational speed (rotation per minute or rpm) of the drive is much slower. This change in the rotational speed ensures the constant flow of data.

CAV (Constant Angular Velocity) on the other hand follows a different method of reading the data. Here the rpm of the motor spinning the media remains constant, regardless of where the data is stored on the media. This makes the drive using this method slower than the drive using the CLV method. The advantage of this method is that it's simple to design and produce the drive because the speed of the motor is constant and latency time (time wasted while the drive changes its rpm) is eliminated.

assortment of 10 files totalling 100 MB was transferred to the hard disk from the CD drive, the **Krypton 52x** clocked 22.21 seconds to complete the task. The slowest drive here turned out to be the **Samsung SC-152**, which took 40.13 seconds to copy the 100 MB file and 34.57 seconds to copy the assorted files.

The most attractive part about this drive is its economical price tag. The runner up in this test was the **50x ASUS CD-S500**, which delivered results that were just a shade below the **Krypton** drive.

The next real-world test that we ran involved opening an assortment of 10 TIFF images in **Photoshop 6.0** from the CD-ROM drive. Since many Web designers, photographers and art professionals extensively use CDs to store backups of images, this test was intended to be indicative of the